

The Algebraic Geometry of Tensor Decomposition

Rank, Symmetry, and Varieties

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What is a tensor?

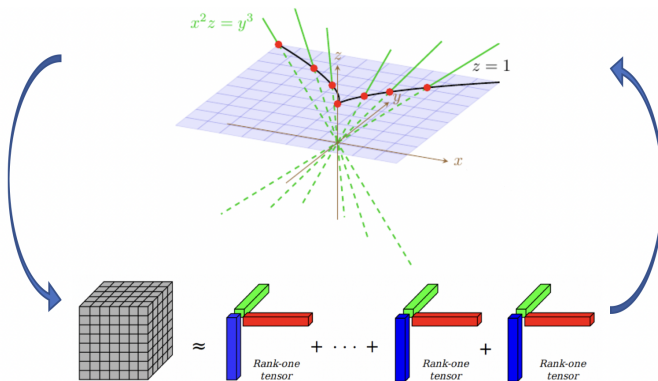
A tensor is a multi-way extension of a matrix.

Tensors can be interpreted as:

- multi-dimensional arrays
- multi-linear maps.

They can represent various types of data, and they are foundational to many fields, such as physics, machine learning, scientific computing, data science and deep learning.

What is the aim of this talk?



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An order- k tensor is a k -dimensional box:

$$A = (a_{i_1, \dots, i_k})_{\{i_j=1, \dots, n_j; j=1, \dots, k\}} \in \mathbb{C}^{n_1 \times \dots \times n_k}$$

There exist many bilinear maps associated to an order- k tensor (after having fixed basis), for example:

$$f_A : \mathbb{C}^{n_1} \times \dots \times \mathbb{C}^{n_{k-1}} \rightarrow \mathbb{C}^{n_k}$$

So A can be thought as an element, for example, of

$$\text{Hom}(\mathbb{C}^{n_1} \otimes \dots \otimes \mathbb{C}^{n_{k-1}}, \mathbb{C}^{n_k})$$

Let's define for now

$$\mathbb{C}^{n_1} \otimes \dots \otimes \mathbb{C}^{n_k} := \text{Hom}(\mathbb{C}^{n_1} \otimes \dots \otimes \mathbb{C}^{n_{k-1}}, \mathbb{C}^{n_k})$$

Tensor Product

We now want to give a more formal definition of tensors, using the **tensor product**. Let us consider 2 vector spaces, V and W , over the field $\mathbb{C}$.

The tensor product

$$\otimes : V \times W \rightarrow V \otimes W$$

is the *unique bilinear map* that satisfies the **universal property**:

- ① $V \otimes W = \text{span}(\otimes(V \times W))$.
- ② If ϕ is a bilinear map $\phi : V \times W \rightarrow H$ then there exists a *unique linear map* f such that the following diagram commutes:

$$\begin{array}{ccc} V \times W & \xrightarrow{\phi} & H \\ \otimes \downarrow & \nearrow f & \\ V \otimes W & & \end{array} \quad \phi(\mathbf{x}, \mathbf{y}) = (f \circ \otimes)(\mathbf{x}, \mathbf{y}).$$

An element of the form

$$\otimes(\mathbf{v}, \mathbf{w}) := \mathbf{v} \otimes \mathbf{w} \in V \otimes W$$

is called a **pure** or **simple tensor** (of order 2). The set of pure tensors is the image of $\otimes$. It will be denoted by $\mathcal{VS}_{m,n} := \text{Im}(\otimes)$ where $m = \dim V$ and $n = \dim W$. The vector space $V \otimes W$ is called the **tensor product** of V and W and its elements are called **tensors**.

Q:

Does the set $\mathcal{VS}_{m,n}$ have any interesting properties/geometric structure?
We're coming back to this later...

Since $V \otimes W$ is a vector space, there exists a basis spanning the space. In fact, it is possible to show that:

Proposition

Let $\{\mathbf{e}_i\}_{i=1}^m$ and $\{\mathbf{f}_j\}_{j=1}^n$ be bases of V and W respectively. Then, $B = \{\mathbf{e}_i \otimes \mathbf{f}_j\}_{i,j=1}^{m,n}$ is a basis of $V \otimes W$.

Hence:

Corollary

If $\dim(V) = m$ and $\dim(W) = n$, $\dim(V \otimes W) = mn$.

After a choice of bases $\{\mathbf{e}_i\}_{i=1}^m$ and $\{\mathbf{f}_j\}_{j=1}^n$ of V and W respectively, the elements $A \in V \otimes W$ can always be represented in coordinates as a 2-dimensional array:

$$(\mathbf{e}_i \otimes \mathbf{f}_j)^* \left(\sum_{p=1}^m \sum_{q=1}^n a_{p,q} \mathbf{e}_p \otimes \mathbf{f}_q \right) = a_{i,j}$$

where

$$(\mathbf{e}_i \otimes \mathbf{f}_j)^* (\mathbf{e}_p \otimes \mathbf{f}_q) = 1, \text{ if } p = i \text{ and } q = j,$$

$$(\mathbf{e}_i \otimes \mathbf{f}_j)^* (\mathbf{e}_p \otimes \mathbf{f}_q) = 0, \text{ if } p \neq i \text{ or } q \neq j.$$

Therefore:

$$\mathcal{A} = \sum_{i=1}^m \sum_{j=1}^n \underbrace{(\mathbf{e}_i \otimes \mathbf{f}_j)^* (\mathcal{A})}_{a_{i,j}} \cdot (\mathbf{e}_i \otimes \mathbf{f}_j).$$

Hence, the 2-dimensional array (i.e. a matrix)

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}$$

represents $\mathcal{A}$ in the basis $B = \{\mathbf{e}_i \otimes \mathbf{f}_j\}_{i,j=1}^{m,n}$.

Corollary

If $V = \mathbb{C}^m$ and $W = \mathbb{C}^n$, then naturally:

$$V \otimes W = \mathbb{C}^m \otimes \mathbb{C}^n \simeq \mathbb{C}^{m \times n} = \mathbb{C}^{mn}.$$

And, of course, **the same holds for more than 2 vector spaces.**

We have seen that

$$V_1 \otimes \cdots \otimes V_d = \text{span}(\otimes (V_1 \times \cdots \times V_d)).$$

$$\dim(V_1 \otimes \cdots \otimes V_d) = n_1 \cdots n_d \neq n_1 + \cdots + n_d = \dim(V_1 \times \cdots \times V_d)$$

$$\begin{array}{ccc} V_1 \times \cdots \times V_d & \xrightarrow{\phi} & H \\ \otimes \downarrow & \nearrow f & \\ V_1 \otimes \cdots \otimes V_d & & \end{array}$$

$\otimes$ is not a bijection over $V_1 \otimes \cdots \otimes V_d$.

Therefore: not every tensor is a pure (or simple) tensor.

For a generic d -tensor $A \in V_1 \otimes \cdots \otimes V_d$, we have:

$$A = \sum_{i=1}^k \alpha_i \mathbf{v}_i^1 \otimes \cdots \otimes \mathbf{v}_i^d.$$

Note that since $\alpha_i \mathbf{v}_i^j \in V_j$ (because they are vector spaces), we can also write

$$A = \sum_{i=1}^k \mathbf{x}_i^1 \otimes \cdots \otimes \mathbf{x}_i^d \text{ without coefficients.}$$

Rank and Border Rank

The **tensor rank decomposition** (TRD) writes a tensor

$$\mathcal{A} \in \mathbb{C}^{n_1} \otimes \cdots \otimes \mathbb{C}^{n_d}$$

as a minimum-length linear combination of simple tensors:

$$\mathcal{A} = \sum_{i=1}^r \alpha_i \mathbf{a}_i^1 \otimes \cdots \otimes \mathbf{a}_i^d, \quad \mathbf{a}_i^k \in \mathbb{C}^{n_k}, \quad \alpha_i \in \mathbb{C}.$$

Definition

The least integer for which $\mathcal{A}$ admits such a decomposition is called the **rank** of $\mathcal{A}$. It is denoted by $\text{rank}(\mathcal{A})$.

A simple tensor is thus by definition a **rank-1 tensor**.

Let us indicate the set of tensors of rank bounded by r with $\mathcal{VS}_{\leq r}$. With the definitions above we can say that the **rank** of $\mathcal{A}$ is the smallest r such that $\mathcal{A} \in \mathcal{VS}_{\leq r}$. On the other hand:

Definition

The **border rank** of $\mathcal{A}$ is defined as the smallest r for which it holds that $\mathcal{A} \in \overline{\mathcal{VS}_{\leq r}}$. We denote it by $\underline{\text{rank}}(\mathcal{A})$.

Obviously,

$$\underline{\text{rank}}(\mathcal{A}) \leq \text{rank}(\mathcal{A}).$$

Rank and Border Rank

Some results:

Proposition

For $\mathbb{C}^{n_1 \times n_2}$, $\overline{\mathcal{V}S_{\leq r}} = \mathcal{V}S_{\leq r}$, i.e. for a matrix $M \in \mathbb{C}^{n_1 \times n_2}$:
 $\text{rank}(M) = \text{rank}(M)$.

Proposition

$$\overline{\mathcal{V}S_{\leq 1}} = \mathcal{V}S_{\leq 1}$$

However, for $r > 1$ border rank and rank can be different!

For example, consider the following sequence of rank-2 tensors

$$\mathcal{A} = \lim_{t \rightarrow \infty} \mathcal{A}_t = \lim_{t \rightarrow \infty} t \left(\mathbf{a}^1 + \frac{1}{t} \mathbf{b}^1 \right) \otimes \left(\mathbf{a}^2 + \frac{1}{t} \mathbf{b}^2 \right) \otimes \left(\mathbf{a}^3 + \frac{1}{t} \mathbf{b}^3 \right) - t \mathbf{a}^1 \otimes \mathbf{a}^2 \otimes \mathbf{a}^3$$

where $\mathbf{a}^k, \mathbf{b}^k \in \mathbb{C}^{n_k}$ are linearly independent, so that $\text{rank}(\mathcal{A}_t) = 2$.

The limit of this sequence is

$$\mathcal{A} = \mathbf{b}^1 \otimes \mathbf{a}^2 \otimes \mathbf{a}^3 + \mathbf{a}^1 \otimes \mathbf{b}^2 \otimes \mathbf{a}^3 + \mathbf{a}^1 \otimes \mathbf{a}^2 \otimes \mathbf{b}^3$$

which is a tensor of rank 3.

In other words, $\overline{\mathcal{V}S_{\leq 2}} \neq \mathcal{V}S_{\leq 2}$ for third-order tensors.

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We have seen several definitions and constructions, but so far geometry does not seem to be involved...

Q:

What if we could look at tensors from another prospective?

What if we could consider tensors as points of a *geometric object*?

By doing so, we could study these geometric objects using all the tools we have from geometric theories, gaining insights on the properties of a given tensor!

In fact: **WE CAN!**

$$\begin{aligned} S_n : \mathbb{P}(V_1) \times \dots \times \mathbb{P}(V_n) &\longrightarrow \mathbb{P}(V_1 \otimes \dots \otimes V_n) \\ ([\mathbf{v}_1], \dots, [\mathbf{v}_n]) &\longmapsto [\mathbf{v}_1 \otimes \dots \otimes \mathbf{v}_n] \end{aligned}$$

$\mathcal{S}_n := \text{Im}(S_n) = \text{rank-1 tensors of order } n$.

Theorem

The set $\mathcal{S}_n$ is indeed an algebraic variety, called **Segre Variety**.

In principle, to show that $\mathcal{S}_n$ is an algebraic variety, we have to find a (homogeneous) ideal $I \in \mathbb{C}[x_0, \dots, x_k]$ such that $V(I) = \mathcal{S}_n$. In other words, we have to find (a finite number of) equations that vanish exactly on $\mathcal{S}_n$.

Luckily for us, finding these equations is rather simple. (We are not going to prove that these are the equations we're looking for...)

$I(\mathcal{S}_n) = ((2 \times 2) - \text{minors of the flattenings of a generic tensor of order } n)$.

Let's look at an example to clarify the computation.

Example

$$\begin{aligned} S_2 : \mathbb{P}(\mathbb{C}^2) \times \mathbb{P}(\mathbb{C}^2) &\longrightarrow \mathbb{P}(\mathbb{C}^2 \otimes \mathbb{C}^2) \simeq \mathbb{P}(\mathbb{C}^4) = \mathbb{P}_{\mathbb{C}}^3 \\ ([x_0, x_1], [y_0, y_1]) &\longmapsto [(x_0, x_1) \otimes (y_0, y_1)] \end{aligned}$$

Let us consider projective coordinates z_0, z_1, z_2, z_3 in $\mathbb{P}_{\mathbb{C}}^3$.

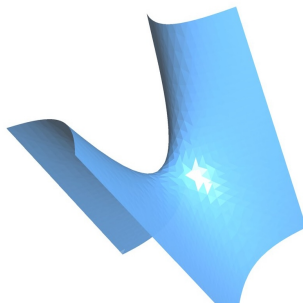
The generic tensor $\mathcal{T} \in \mathbb{C}^2 \otimes \mathbb{C}^2$ writes as:

$$\mathcal{T} = \begin{bmatrix} z_0 & z_1 \\ z_2 & z_3 \end{bmatrix}$$

Hence, rank-1 tensors of $\mathbb{C}^2 \otimes \mathbb{C}^2$ can be seen as points of the quadric $X \in \mathbb{P}_{\mathbb{C}}^3$ identified by the polynomial

$$z_0 z_3 - z_1 z_2.$$

In the affine chart $U_3 = \{z_3 \neq 0\}$, these tensors correspond to the points of $\mathbb{C}^3$ that lie on the surface: $X_3 : x - yz = 0$.



Example

Let us now consider 2 distinct points of X , say A and B .

Without loss of generality, we can consider $A, B \in X_3$.

We know that these 2 points have an interpretation as rank-1 tensors.

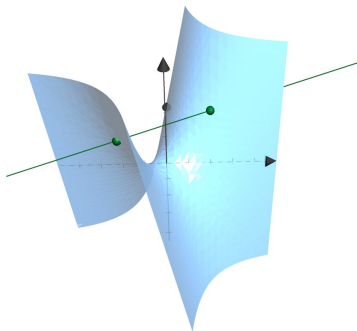
Now, consider $(\lambda, \mu) \neq (0, 0)$. The (projective) point

$$\mathcal{T}' = \lambda A + \mu B$$

is a point of the (projective) line through A and B .

But it can be also seen as a rank-2 tensor!

In general, the points of a line through two distinct points of the Segre Variety have a natural interpretation as rank-2 tensors.



$\mathcal{T}$ has rank $\leq r \iff \exists P_1, \dots, P_r \in \mathcal{S}_n$ such that $\mathcal{T} \in \langle P_1, \dots, P_r \rangle$.

Definition

The set of tensors of rank $\leq r$ is:

$$\sigma_r^o(\mathcal{S}_n) := \bigcup_{P_1, \dots, P_r \in \mathcal{S}_n} \langle P_1, \dots, P_r \rangle.$$

Problem: this set is not an algebraic variety (it is not Zariski-closed!). However, if we take the closure, we obtain...

Definition

The r^{th} -secant variety of $\mathcal{S}_n$ is: $\sigma_r(\mathcal{S}_n) = \overline{\sigma_r^o(\mathcal{S}_n)}$

Asking for our set to be a variety, we gain geometric structure but we lose our previous algebraic interpretation.

In fact, algebraically, the points of $\sigma_r(\mathcal{S}_n)$ can be seen as **tensors of border rank $\leq r$** .

Q:

Can we find equations of the Secant Variety?

This is still an open problem, however there is a case where we are able to say more:

Symmetric Tensors

$$\begin{aligned} \pi_s : V^{\otimes d} &\longrightarrow V^{\otimes d} \\ v_1 \otimes \dots \otimes v_d &\longrightarrow \frac{1}{d!} \sum_{\sigma \in S_d} v_{\sigma(1)} \otimes \dots \otimes v_{\sigma(d)} \end{aligned}$$

$S^d(V) := \text{Im}(\pi_s)$ is the subspace of $V^{\otimes d}$ of symmetric d -tensors.

In coordinates (after fixing a basis), symmetric tensors are hypermatrixes that are symmetric in the entries.

Now, let us consider the following map:

$$\begin{aligned} \mathcal{V}_d : \mathbb{P}(V) &\longrightarrow \mathbb{P}(S^d(V)) \\ [v] &\longmapsto [v^{\otimes d}] \end{aligned}$$

Again, the space $\mathcal{V}_d := \text{Im}(\mathcal{V}_d)$ is an algebraic variety called the **Veronese Variety**.

The points of the Veronese Variety have a natural interpretation as rank-1 **symmetric** tensors.

As before:

$$I(\mathcal{V}_d) = ((2 \times 2) - \text{minors of the flattenings of a generic symmetric tensor of order } d).$$

$$\begin{aligned}\mathcal{V}_2 : \mathbb{P}(\mathbb{C}^2) &\longrightarrow \mathbb{P}(S^d(\mathbb{C}^2)) \subset \mathbb{P}(\mathbb{C}^4) = \mathbb{P}_{\mathbb{C}}^3 \\ [x_0, x_1] &\longmapsto [(x_0, x_1) \otimes (x_0, x_1)]\end{aligned}$$

Let us consider projective coordinates z_0, z_1, z_2, z_3 in $\mathbb{P}_{\mathbb{C}}^3$.

The generic tensor $\mathcal{T} \in \mathbb{C}^2 \otimes \mathbb{C}^2$ writes as:

$$\mathcal{T} = \begin{bmatrix} z_0 & z_1 \\ z_2 & z_3 \end{bmatrix}$$

We are asking for $\mathcal{T}$ to be symmetric, therefore we are working on the hyperplane $H : z_2 = z_1$. Hence, rank-1 symmetric tensors of $\mathbb{C}^2 \otimes \mathbb{C}^2$ can be seen as points of the quadric $X \in \mathbb{P}_{\mathbb{C}}^3$ identified by the polynomial

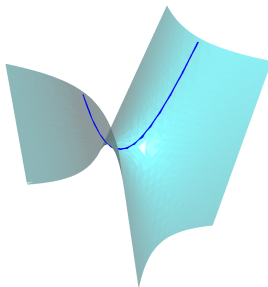
$$z_0 z_3 - z_1 z_2$$

on the hyperplane H .

In other words, these are the points of the variety:

$$Y = \begin{cases} z_0 z_3 - z_1^2 = 0 \\ z_1 = z_2 \end{cases}$$

In the affine chart $U_3 = \{z_3 \neq 0\}$, these tensors correspond to the points of $\mathbb{C}^3$ that lie on the surface: $X_3 : x - yz = 0$ and on the plane $y = z$.



As expected, our Veronese Variety (in dark blue) is a subvariety of our Segre Variety. Indeed, every rank-1 symmetric tensor is obviously a rank-1 tensor! We can do more... In general, we have an algebraic characterization of symmetric tensors, through the following:

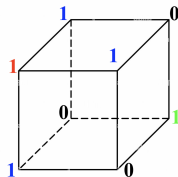
Preposition

Symmetric tensors of higher order $= d$ are in bijection with homogeneous polynomials of degree d

$$\mathbb{C}[x_0, \dots, x_n]_d \longrightarrow S^d(\mathbb{C}^{n+1})$$

$$f = \sum_{|\underline{\alpha}|=d} a_{\underline{\alpha}} \mathbf{x}^{\underline{\alpha}} \longmapsto \left(\binom{d}{\underline{\alpha}}^{-1} a_{\underline{\alpha}} \right)$$

$$x^3 + 3x^2y + y^3 \mapsto$$



$$e_0 \otimes e_0 \otimes e_0 + e_0 \otimes e_0 \otimes e_1 + e_0 \otimes e_1 \otimes e_0 + e_1 \otimes e_0 \otimes e_0 + e_1 \otimes e_1 \otimes e_1$$

Working with symmetric tensors is equivalent to working with homogeneous polynomials!

Note that in general, rank-1 symmetric tensors correspond to power of linear forms, through the bijection seen above:

$$v^{\otimes d} = [a_1, \dots, a_n]^{\otimes d} \mapsto (a_0x_0 + \dots + a_nx_n)^d$$

The introduction of symmetric tensors leads us to another definition:

Definition

The **symmetric rank** is the minimum r such that a symmetric tensor $\mathcal{T}$ can be written as a sum of rank-1 symmetric tensors:

$$\mathcal{T} = \sum_{i=1}^r v_i^{\otimes d}$$

Equivalently, working with polynomials, we have:

Definition

The **symmetric rank** of a polynomial F is the minimum r such that:

$$F = \sum_{i=1}^r L_i^d, \quad L_i = \text{linear forms}$$

From now on, we are focusing on *computing the symmetric rank of a symmetric tensor* or, equivalently, we are focusing on *computing the symmetric rank of a polynomial*.

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Recall:

$I(\mathcal{V}_d) = ((2 \times 2) - \text{minors of the flattenings of a generic symmetric tensor of order } d)$.

We also have another characterization of those minors $\longrightarrow$ **Catalecticant Maps**.

Definition

$F \in S^d(\mathbb{C}^n)$, $0 \leq i \leq d$. The i^{th} catalecticant map associated to F is:

$$C_F^{i,d-i} : \mathbb{C} \left[\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_n} \right]_i \longrightarrow \mathbb{C}[x_1, \dots, x_n]_{d-i}$$
$$\partial_{\mathbf{x}} \longmapsto \partial_{\mathbf{x}}(F)$$

The matrix associated to $C_F^{i,d-i}$ is called **catalecticant matrix**. All these matrixes have different sizes in dependence of i .

The surprising part is that the first catalecticant matrix ($i = 1$) of F is the standard 1^{st} flattening of the symmetric tensor $\mathcal{T}$ corresponding to F (modulo some constants that depend on the basis choice).

Hence, we have an algorithm to test if a symmetric tensor has rank 1.

Example

Given an $F \in S^d(V)$, we have to:

1) Compute the rank of $C_F^{1,d-1}$

1.1) If $\text{rank}(C_F^{1,d-1}) = 1$, then $\text{rank}(F) = 1$

1.2) Otherwise $\text{rank}(F) > 1$.

Example 1

$$F = (x + y + z)^3.$$

$$C_F^{1,2} : \mathbb{C} \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right]_1 \longrightarrow \mathbb{C}[x, y, z]_2$$
$$\begin{aligned} \frac{\partial}{\partial x} &\longmapsto 3(x + y + z)^2 \\ \frac{\partial}{\partial y} &\longmapsto 3(x + y + z)^2 \\ \frac{\partial}{\partial z} &\longmapsto 3(x + y + z)^2 \end{aligned}$$

$$C_F^{1,2} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad \text{rank}(C_F^{1,2}) = 1.$$

Example 2

$$F = x^3 + y^3 + z^3.$$

$$C_F^{1,2} : \mathbb{C} \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right]_1 \longrightarrow \mathbb{C}[x, y, z]_2$$
$$\begin{aligned} \frac{\partial}{\partial x} &\longmapsto 3x^2 \\ \frac{\partial}{\partial y} &\longmapsto 3y^2 \\ \frac{\partial}{\partial z} &\longmapsto 3z^2 \end{aligned}$$

$$C_F^{1,2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \text{rank}(C_F^{1,2}) \neq 1.$$

The (2×2) -minors of a catalecticant matrix characterize rank-1 polynomials.

Q:

What about the (3×3) -minors? Is there any relation with the rank-2 polynomials?

Let us now focus on $F \in \mathbb{C}[x, y]_d = S^d(\mathbb{C}^2)$.

1st Catalecticant

$$F = x^4 + y^4.$$

$$C_F^{1,3} : \mathbb{C} \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right]_1 \longrightarrow \mathbb{C}[x, y, z]_3$$

$$\frac{\partial}{\partial x} \longmapsto 4x^3$$

$$\frac{\partial}{\partial y} \longmapsto 4y^3$$

$$C_F^{1,3} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad \text{rank}(C_F^{1,3}) = 2.$$

2nd Catalecticant

$$F = x^4 + y^4.$$

$$C_F^{2,2} : \mathbb{C} \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y} \right]_2 \longrightarrow \mathbb{C}[x, y, z]_2$$

$$\frac{\partial^2}{\partial x^2} \longmapsto 12x^2$$

$$\frac{\partial^2}{\partial x \partial y} \longmapsto 0$$

$$\frac{\partial^2}{\partial y^2} \longmapsto 12y^2$$

$$C_F^{2,2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \text{rank}(C_F^{2,2}) = 2.$$

Apolarity Lemma and Related Results

Fact 1: The existence of higher order minors is just a matter of size.

Fact 2: The rank of the catalecticant stabilizes as soon as the size is big enough.

Fact 3: For 2-variables polynomials, **the most squared catalecticant can be used to characterize the border rank.**

To prove these results, we have to use another important tool: the **apolarity lemma**.

Apolarity Lemma

$$F = \xi_1 L_1^d + \dots + \xi_r L_r^d \iff I([L_1], \dots, [L_r]) \subseteq F^\perp = \{\partial : \partial F = 0\} = \bigcup_i \ker C_F^{i, d-i}.$$

We are going to skip all the details, but we mention the following crucial consequence:

Theorem

In the 2 variable case, $I(\sigma_r(\mathcal{V}_d)) = ((r+1)\text{-minors of the most squared catalecticant})$.

We can also choose the "smallest" catalecticant where the $r+1$ -minors can be computed. The following holds:

Theorem

Let $F \in S^d(\mathbb{C}^2)$ be of border rank $= r$. Then $\text{rank}(F) \in \{r, d - r + 2\}$.

Sylvester's Algorithm

- ✓ The rank of a catalecticant tells us in which $\sigma_r(\mathcal{V}_d)$ a given F belongs.
- ✓ How can we distinguish if F either has rank r or if $F \in \sigma_r(\mathcal{V}_d) \setminus \sigma_r^o(\mathcal{V}_d)$? It is sufficient to look if $\exists r$ distinct points on the apolar.
- ✓ How to compute the rank and a decomposition of such F ? We use the points on the apolar.

Sylvester's Algorithm

Input: $F \in S^d(\mathbb{C}^2)$.

Output: rank, border rank, decomposition of F .

① Compute the rank of $C_F^{d/2, d/2} := r$

② Compute $\ker(C_F^{r, d-r}) =: \mathcal{K}$

- if a generic element of $\mathcal{K}$ has r distinct roots $L_1, \dots, L_r$, then $\text{rank}(F) = r$ and:

$$F = \sum_{i=1}^r \xi_i L_i$$

- otherwise $\text{rank}(F) = d - r + 2$.

③ Compute the generic element of $\ker(C_F^{d-r+2, r-2}) =: \tilde{\mathcal{K}}$. Its distinct roots $L_1, \dots, L_{d-r+2}$ give the decomposition:

$$\sum_{i=1}^{d-r+2} \xi_i L_i$$

$$F = x^2 y^4$$

$$1) C_F^{3,3} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 3 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \Rightarrow \text{rank}(C_F^{3,3}) = 3 \Rightarrow \underline{\text{rank}}(F) = 3$$

Therefore:

$$\text{rank}(F) = \begin{cases} 3 \\ d - r + 2 = 5 \end{cases}$$

$$2) \ker(C_F^{3,3}) = \langle (1, 0, 0, 0) \rangle \subset S^3(\mathbb{C}^2)$$

$$\ker(C_F^{3,3}) \ni u = u_0^3 \longrightarrow \text{NO 3 distinct roots} \longrightarrow \text{rank}(F) = 5.$$

$$3) C_F^{5,1} = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \end{pmatrix}$$

$$\ker(C_F^{5,1}) \ni u = u_0^5 + u_0^4 u_1 + u_0^3 u_1^2 + u_1^5 \dots \longrightarrow \text{Numerically I can find 5 distinct roots.}$$

1 Introduction

2 Basic Definitions

- An Intuition on Tensors
- Tensor Product
- Rank and Border Rank

3 Where is the Algebraic Geometry?

- The Segre Variety
- Secant Variety
- Symmetric Tensors and the Veronese Variety

4 Computing the Rank

- Catalecticant Maps
- Apolarity Lemma and Related Results
- Sylvester's Algorithm

5 Some Applications

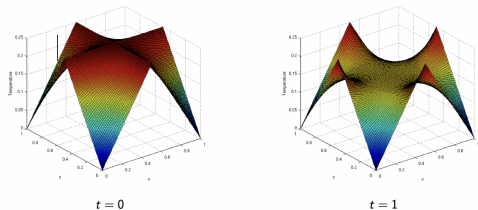
6 References

Because each rank-1 summand in a tensor rank decomposition of a tensor is usually unique and always easy to interpret, the tensor rank decomposition is often used as a tool for **explorative data analysis** or a **reduced order model** of (potentially incomplete) multidimensional data.

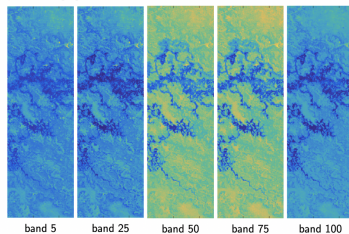
- **Factor analysis** for analyzing underlying causes in psychology (*Carroll and Chang, 1970; Harshman, 1970*)
- Identifying spectra of fluorophores in **fluorescence spectroscopy** (*Appellof and Davidson, 1981*)
- **Estimation of the parameters of some latent variable models** like single exchangeable topic models, mixtures of Gaussians, and hidden Markov models (*Anandkumar et al., 2014*)
- Tensor-based **collaborative filtering** and **recommender systems** (*Frolov and Oseledets, 2017*)
- Exploratory analysis in **sports analytics** applications (*Verstraete, Decroos, Coussement, Vannieuwenhoven and Davis, 2020; Geens, 2020*)
- Reduced order model for **thermodynamic quantities of multicomponent alloys** (*Coutinho, Vervliet, De Lathauwer and Moelans, 2020*)
- Reduced order model for **power output of curved solar panels** (*De Coppel, 2021*)
- Reduced order model for finite element matrices of **parameterized joint elements in lattices** (*De Weer, Vannieuwenhoven, Lammens, Meerbergen, 2021*)

Some Applications

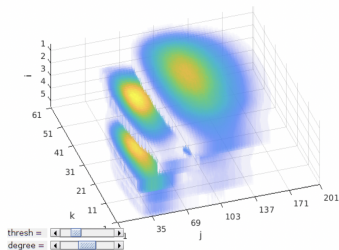
(1) Solutions of (parameterized) partial differential equations



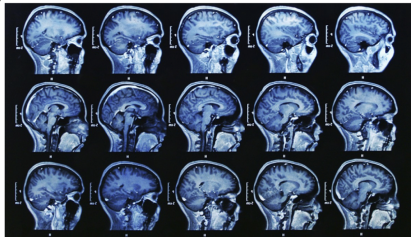
(2) hyperspectral imaging



(3) fluorescence spectroscopy



(4) MRI data



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5 Some Applications

6 References

- ◇ *Tensor Decomposition for Big Data Analysis*, course of the Master in Mathematics of the University of Trento, held by Professor **Alessandra Bernardi** (University of Trento).
- ◇ *Tensor decompositions and their applications*, masterclass held by Professor Nick Vannieuwenhoven (KU Leuven).
- ◇ *Tensors: Geometry and Applications*, J.M. Landsberg.
- ◇ *Symmetric Tensor Decomposition*, J. Brachat et al.

THANK YOU FOR YOUR ATTENTION!

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Check out the **TensorDec Laboratory** of UniTrento:

